Persona Vectors as a Psychometric Instrument for Moral Character in Language Models

A research proposal

Manou Rimer · BlueDot Impact, Technical AI Safety · June 2026

Project summary

This project builds on *Persona Vectors* (Chen et al., 2025) and the authors' `safety-research/persona_vectors` codebase, which extract linear directions in a model's activation space whose projection predicts the character traits—honesty, harmfulness, sycophancy—a model will go on to express. I treat that projection as a *measurement instrument* for a model's moral character and ask the two questions any instrument must pass before it can be trusted: (1) **structure**—when a model behaves ethically, is that driven by a single latent "morality" dimension or by several separable ones (honesty, harm-avoidance, fairness) that can move independently?—and (2) **reliability**—does the reading stay stable when the same moral situation is reworded or reframed, or does it track surface features of the prompt rather than the underlying disposition? Using Gemma-3-12B, I extract and validate persona vectors for a small set of a-priori-separable moral traits (honesty, harmlessness, fairness, compassion), then apply factor analysis (for structure) and generalizability theory (for reliability) to the resulting projections. This first pass is scoped as a minimum viable study on stated moral judgment; the predictive "knowing–doing" question and agentic behavior are deferred to a later extension.

Motivation

A central bet in AI safety is that we can monitor a model's internal states to catch misaligned dispositions—deception, harmfulness—before they surface in behavior, rather than relying on outputs alone. Persona vectors make this concrete: a model leaves an internal "tell" that can be read before it produces a single token, and the strength of that reading roughly tracks how the model subsequently behaves. The appeal for oversight is obvious—an internal gauge of a model's moral state would let us watch for drift and intervene early.

But a gauge is only useful if it measures the right quantity and gives a steady reading. Before a persona-vector projection can be treated as a measure of moral character, two psychometric properties must hold, neither of which has been established. The first is *dimensionality*: if honesty, fairness, and harm-avoidance rise and fall together, a single "moral compass" gauge suffices; if they move independently, monitoring requires several gauges, and "making a model more ethical" is less like turning one knob than balancing an equalizer. The second is *reliability*: a thermometer that reports a different temperature depending on how you hold it is worthless. If the projection jumps when a moral question is merely rephrased, it is reading the prompt, not the disposition. This project checks the instrument itself before anyone leans on it—the unglamorous groundwork that has to precede interpretability-based moral monitoring. Crucially, both outcomes are informative: an instrument that proves unreliable is itself a cautionary finding for the field, not a null result.

Related work

Persona vectors and activation directions. Chen et al. (2025) introduce an automated pipeline that turns a natural-language description of a trait into a linear direction in activation space, and show that the projection of the final prompt-token activation onto that direction predicts trait expression, enabling monitoring and steering. This sits in a broader line of work finding behaviorally meaningful linear directions in the residual stream—e.g. Arditi et al. (2024), who show refusal is mediated by a single direction. These results establish that directions *exist and predict*; they do not characterize the geometry of the trait space or test the stability of the readings, which is the gap this project addresses.

Moral evaluation benchmarks. The ETHICS dataset (Hendrycks et al., 2021) supplies moral-judgment items spanning justice, deontology, virtue, utilitarianism, and commonsense morality, and serves as the item bank for this first pass. Moral Foundations Theory (Graham et al.) is taken as *inspiration* rather than a framework to reproduce: its claim that morality is plural—several distinct concerns rather than one—motivates choosing traits that span different moral domains so that the dimensionality question stays genuinely open. A later extension can add agentic behavior benchmarks such as MACHIAVELLI (Pan et al., 2023) to separate what a model states is right from what it does.

Psychometric measurement theory. The questions here are old ones in disguise. Factor analysis asks how many latent dimensions explain a set of correlated observations—exactly the "one moral axis or several" question. Generalizability theory (Cronbach et al., 1972; Brennan, 2001) decomposes the variance in a measurement into the signal of interest versus the contribution of incidental conditions (here: phrasing and framing), yielding a single coefficient for how dependable a reading is. This project transplants both onto activation-space projections.

Research questions

RQ1 (structure). Is moral behavior, as read through persona-vector projections, governed by one latent dimension or several separable ones? The original authors explicitly raise whether a natural "persona basis" exists and whether correlations between persona vectors predict co-expression of traits, but never resolve it—and never for moral traits.

RQ2 (reliability). Is the persona-vector projection a dependable measurement of a moral trait under incidental variation in how the situation is presented? This falls outside even the paper's stated open problems: its monitoring results presume the projection is dependable without testing it.

Method — experimental pipeline

Stage 0 — Define the construct space. Select a small set of morally relevant traits to extract vectors for: **honesty, harmlessness, fairness, and compassion**. These are chosen (with Moral Foundations Theory as inspiration) to span distinct moral concerns—truth, harm, justice, and care—so that either a one-factor or a multi-factor result is genuinely possible rather than baked in by the choice of near-synonyms. Each also admits a clear contrastive description, which tends to make for cleaner extraction than more abstract traits such as sanctity. Fix the model (Gemma-3-12B) and a target layer (swept later).

Stage 1 — Extract and validate the vectors. Using the persona_vectors pipeline, derive each trait vector by contrasting activations under trait-expressing vs. trait-suppressing prompts (difference of means at the last prompt token). Validate each vector before use: its projection must cleanly separate held-out expressing vs. suppressing prompts, and steering along it must shift behavior. Traits whose vectors fail to validate are flagged—itself a small finding bearing on whether some moral traits are less accessible to linear methods.

Stage 2 — Build the item bank. Assemble roughly 300 moral-judgment items from ETHICS, each tagged by the trait it most engages. Generate about three paraphrase variants per item and define the measurement conditions (facets) for the reliability study: paraphrase and framing. (Sampling seed is intentionally excluded—the projection is read from a deterministic forward pass at the last prompt token, so seed would introduce no variance in the measurement; it only matters once sampled behavior is added in a later pass.)

Stage 3 — Harvest the projection data. The single GPU-bound step. For every item under every condition, run a forward pass and cache the layer-L, last-token residual-stream activation; the projection onto each trait vector is then cheap, re-runnable CPU work. Caching the raw activations (not just the projections) means new traits or layers never require re-harvesting. The model's stated judgment for each item is recorded alongside, for descriptive checks and to seed the later predictive extension.

Stage 4 — Dimensionality (factor analysis). Average over the incidental conditions to obtain a clean item-by-trait table of projections, standardize it, and run exploratory factor analysis / PCA with parallel analysis to determine how many independent dimensions are present and which traits cluster. An essential control compares against random or orthogonalized directions, since trait vectors can correlate from shared prompt format rather than shared moral content.

Stage 5 — Reliability (generalizability theory). Treat the item as the object of measurement and paraphrase and framing as facets. Run a crossed design, estimate the variance components with a mixed model, then a decision study to obtain a generalizability coefficient per trait and to answer the practical question: how many paraphrases must be averaged to obtain a dependable reading?

Feasibility and compute

The study is forward-pass only—no model training—so compute is modest. All inference runs on Gemma-3-12B via Modal, where the activation harvest (Stage 3) is the only meaningful GPU cost; a clean end-to-end run is a small number of GPU-hours, with development iteration the larger share. Stages 4–5 are CPU-bound statistical analysis. A minimal viable version—four traits, ETHICS items only, around 300 items, three paraphrases, one layer—already delivers a dimensionality result and a genuine generalizability study, and fits within a short project timeline.

Expected outcomes and interpretation

Each result is informative regardless of direction. On **structure**: a single dominant factor would suggest models carry something like one internal moral compass, simplifying monitoring to a single quantity; a multi-factor solution would show that moral traits are represented separately, so oversight must track several and interventions on one trait may not transfer to others. On **reliability**: a high generalizability coefficient would license building monitoring tools on top of persona-vector readings; a low one would be a substantive cautionary result, indicating that projections respond substantially to surface phrasing and that interpretability-based moral oversight needs more careful measurement than current practice assumes. Establishing either property is a prerequisite for using persona vectors in real moral monitoring.

Selected references

Chen, R., et al. (2025). *Persona Vectors: Monitoring and Controlling Character Traits in Language Models.* arXiv:2507.21509.

Arditi, A., et al. (2024). *Refusal in Language Models Is Mediated by a Single Direction.* arXiv:2406.11717.

Hendrycks, D., et al. (2021). *Aligning AI With Shared Human Values* (the ETHICS dataset). ICLR. arXiv:2008.02275.

Pan, A., et al. (2023). *Do the Rewards Justify the Means? Measuring Trade-offs Towards Ethical Behavior with the MACHIAVELLI Benchmark.* ICML. arXiv:2304.03279.

Graham, J., Haidt, J., et al. *Moral Foundations Theory: The Pragmatic Validity of Moral Pluralism.* Advances in Experimental Social Psychology.

Cronbach, L. J., et al. (1972). *The Dependability of Behavioral Measurements: Theory of Generalizability for Scores and Profiles.* Wiley.

Brennan, R. L. (2001). *Generalizability Theory.* Springer.

Note: arXiv identifiers are provided where known; please verify before formal submission.